DNS Fundamentals for Wall-Bounded Thermal Flows

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Outline

Turbulence
- Turbulence Governing Equations

DNS
- Turbulence Scales

Resolution
- Wall-Bounded Flows

Time Stepping
- CFL and Diffusion Constraints

Special Considerations
- Thermally Developing Flow

Diagnostics
- Checking DNS Quality

Summary
- Key Points

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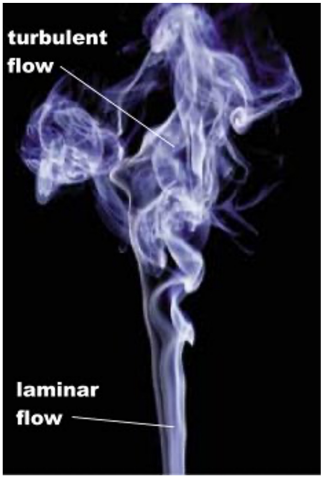
What is Turbulence?

Description, not definition

- **Irregularity in both space and time** (Tennekes and Lumley, 1972; Ueda and Hinze, 1975; Pope, 2001)
- **Non-repetition** (Launder, 1991)
- **Three-dimensional, unsteady eddies with different scales** (Bradshaw, 1971; Tennekes and Lumley, 1972)
- **Diffusivity** (Tennekes and Lumley, 1972; Pope, 2001)
- **Dissipation** (Pope, 2001)
- **Flow property, not a fluid property**
- **Sensitivity to disturbances** (Pope, 2001)



Laminar and turbulent flow

Turbulence Energy Cascade and Spectral Scales

Turbulent kinetic energy spans a wide range of eddy scales and is described by the spectrum $E(\kappa)$.

Integral / large scales

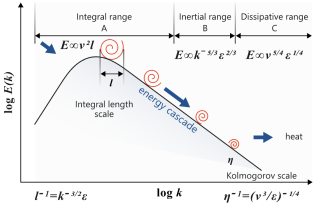
- largest and most energetic eddies
- geometry-dependent and anisotropic
- extract energy from the mean flow

Inertial range

- energy cascades from large to small scales
- driven by vortex stretching and eddy breakdown
- follows $E(\kappa) \sim \kappa^{-5/3}$

Kolmogorov / dissipative scales

- smallest eddies
- viscosity-dominated and nearly isotropic
- dissipate kinetic energy into heat



Turbulence energy spectrum

Governing Equations

For variable-density thermal flows, a conservative form is commonly used.

Continuity, momentum and energy equations

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0$$

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho f_i$$

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(\rho h u_j)}{\partial x_j} = \frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right) + S_h$$

Low-Mach Number Approximation

For low-Mach-number / incompressible variable-property thermal flows, a simplified conservative form is commonly used.

Conservative Variables and Closure

Conservative variables

$\rho, \quad \rho u_i, \quad \rho h$

Thermophysical closure

$$\rho,\ h,\ \mu,\ k = f(T, p_{\mathrm{th}})$$

Alternatively, properties may be obtained from a look-up table.

Typical workflow

$$\rho h \rightarrow T \rightarrow \rho, \mu, k$$

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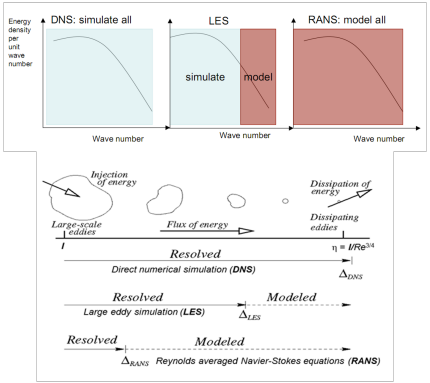
Methods to Solve Turbulence: RANS, LES, DNS

RANS

- Models **all turbulence scales**
- Solves time-averaged equations
- Uses turbulence models (e.g. k–ε, k–ω)
- **Lowest computational cost**

LES (Large Eddy Simulation)

- Resolves **large eddies**
- Models small (subgrid) scales
- Captures unsteady 3D structures
- **Moderate computational cost**



From RANS → LES → DNS: More scales resolved ⇒ higher accuracy ⇒ higher cost

Direct Numerical Simulation (DNS)

What is DNS?

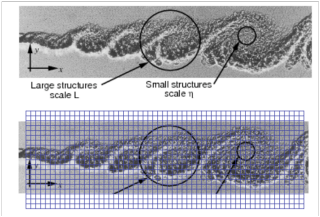
- Direct simulation of the **Navier–Stokes equations**
- **No turbulence modelling assumptions**
- Resolves ALL relevant spatial and temporal scales
  - **In space:** resolves all eddy scales (largest → smallest)
  - **In time:** resolves all relevant time scales

Limitation

Very high computational cost

Why DNS?

- **Highest fidelity**: minimal modelling assumptions
- Captures full multi-scale turbulence physics
- Provides **reference-quality data** for validation and model development



DNS resolving full turbulence spectrum

Kolmogorov Scales

Physical meaning

- Smallest turbulent eddy scales
- Located in the **dissipative range**
- Viscous effects dominate
- Kinetic energy → heat

Key property

Small-scale turbulence is approximately **isotropic**

Kolmogorov scales

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4}$$

$$\tau_\eta = \left(\frac{\nu}{\varepsilon}\right)^{1/2}$$

$$v_\eta = (\nu\varepsilon)^{1/4}$$

Implication for DNS

DNS must resolve down to η and τ_η

Scale Separation with Reynolds Number

Physical picture

- Turbulence spans a wide range of scales

$$l \to \eta$$

- Scale separation increases with Reynolds number

Implication for DNS

Higher $Re \Rightarrow$ more scales to resolve $\Rightarrow$ finer grids & smaller Δt

Scaling laws

Length

$$\frac{l}{\eta} \sim Re^{3/4}$$

Time

$$\frac{t}{\tau_\eta} \sim Re^{1/2}$$

Velocity

$$\frac{v}{v_\eta} \sim Re^{1/4}$$

DNS Cost from Scale Separation

Spatial resolution

- DNS must resolve all scales: $l \to \eta$
- Grid requirement:

$$N \sim \left(\frac{l}{\eta}\right)^3 \sim Re^{9/4}$$

Interpretation

More scale separation $\Rightarrow$ more grid points

Temporal resolution

- Smallest time scales must be resolved:

$$\frac{t}{\tau_\eta} \sim Re^{1/2}$$

Total computational cost

$$\text{Cost}_{DNS} \sim N \times N_t$$
$$\sim Re^{9/4} \times Re^{1/2} = Re^{11/4}$$

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Wall-Bounded Flows: Channel and Pipe DNS

Flow characteristics

- Near-wall turbulence is **highly anisotropic**
- Strong gradients near the wall
- Wall-normal resolution is **critical**

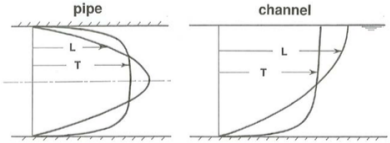
Key Feature

Near-wall physics controls DNS resolution requirements

Wall-unit definitions

$$\tau_w = \mu_w \left.\frac{\partial U}{\partial y}\right|_{wall}, \quad u_\tau = \sqrt{\frac{|\tau_w|}{\rho_w}}$$

$$y^+ = \frac{y u_\tau}{\nu}, \quad U^+ = \frac{U}{u_\tau}, \quad Re_\tau = \frac{u_\tau H}{\nu}$$



DNS Grid Resolution: Channel Flow

Wall-unit resolution

$$\Delta x^+ = \frac{\Delta x u_\tau}{\nu}$$

Typical DNS guidelines

$$\Delta x^+ \approx 8\text{–}12, \quad \Delta z^+ \approx 4\text{–}6, \quad \Delta y^+_{wall} \lesssim 1$$

Domain size

$$L_x^+ \sim O(10^3), \quad L_z^+ \sim O(10^3)$$

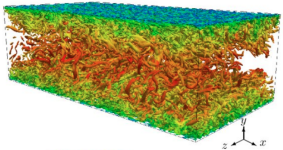
L_z^+ may be $O(10^2)$ only for minimal-box near-wall studies.

Grid characteristics

- Uniform in streamwise and spanwise
- Strong clustering near walls

Implication for DNS

- Accurate near-wall resolution is essential for correct turbulence statistics
- Must resolve near-wall streaks and vortices, buffer-layer turbulence and outer-layer large-scale motions



Representative Channel-Flow DNS Cases

Re_τ	Typical use	Cost	Comment
180	Training / validation	Low	classic benchmark
395	Standard validation	Moderate	widely used reference
590	Higher-Re validation	Moderate–high	reduced low-Re effects
1000–2000	Research	High	clearer scale separation
5200	High-Re DNS	Very high	Lee & Moser-type datasets
10000	State-of-the-art	Extreme	very large HPC calculation

Key message

- $Re_\tau = 180$ and 395 are ideal for code testing and validation.
- Higher Re_τ cases are useful for physics studies but require much larger grids and longer averaging.

DNS Grid Resolution: Pipe Flow

Wall-unit resolution

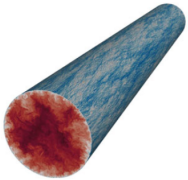
$$\Delta x^+ = \frac{\Delta x u_\tau}{\nu}, \quad \Delta r^+ = \frac{\Delta r u_\tau}{\nu}$$

$$r\Delta\theta^+ = \frac{r\Delta\theta u_\tau}{\nu}$$

Typical DNS guidelines

$$\Delta x^+ \approx 6\text{–}10$$

$$r\Delta\theta^+ \approx 4\text{–}6, \qquad \Delta r^+_{wall} \lesssim 1$$



Grid characteristics

- Axial direction similar to channel flow
- Strong radial clustering near the wall
- Azimuthal spacing depends on radius:

$$\Delta s_\theta = r\Delta\theta$$

Additional challenges

- Coordinate singularity at $r = 0$
- Adequate resolution required in both wall and core regions
- Smooth transition from near-wall grid to centreline

Representative Pipe-Flow DNS Cases

Re_τ	Typical use	Cost	Comment
180–360	Training / low-Re pipe	Low	useful for code testing
550–1000	Standard pipe DNS	Moderate	common validation range
2000	Advanced pipe DNS	High	large-scale motions clearer
5200	State-of-the-art pipe DNS	Very high	recent high-Re DNS level

Key message

- Low-Re pipe cases are suitable for learning, debugging and verification.
- High-Re pipe DNS is more demanding than channel DNS due to cylindrical geometry and centreline treatment.

For the same target Re_τ, pipe DNS often requires additional care because of cylindrical coordinates and centreline treatment.

Thermal DNS: Resolution Requirements

Why thermal DNS is more demanding

- Must resolve both velocity and temperature fields
- Thermal boundary layer depends on:

$$Pr = \frac{\nu}{\alpha}$$

Thermal vs viscous scales

- $Pr > 1$: thinner thermal boundary layer
- $Pr < 1$: more diffusive temperature field

Thermal wall-unit constraint

$$y_T^+ = y^+ Pr$$

$$\Delta y_{wall,T}^+ = \Delta y_{wall}^+ Pr \lesssim 1$$

$$\Rightarrow \Delta y_{wall}^+ \lesssim \frac{1}{Pr}$$

Key message

Thermal DNS may impose stricter near-wall resolution than velocity DNS, especially for high-Pr fluids.
$\Delta y_{wall}^+ \lesssim \frac{1}{Pr}$ is a rough guideline, especially for variable-property fluids.

Outline

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Time Stepping
- CFL and Diffusion Constraints

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Time-Stepping: Channel Flow

Convective stability

$$CFL = \max\left(\frac{|u|\Delta t}{\Delta x}, \frac{|v|\Delta t}{\Delta y}, \frac{|w|\Delta t}{\Delta z}\right) \lesssim 0.5\text{–}1.0$$

Diffusive stability

$$D = \nu\Delta t\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}\right) \lesssim 0.5$$

Typical wall-unit time step

$$\Delta t^+ \approx 0.2\text{–}0.5 \quad (Re_\tau = 180)$$

$$\Delta t^+ \approx 0.15\text{–}0.4 \quad (Re_\tau = 395)$$

Implications

- Near-wall Δy can strongly restrict Δt
- Higher Re_τ often requires smaller physical time steps

Key message

Channel DNS time step is limited by both convection and near-wall diffusion.

Time-Stepping: Pipe Flow

Convective stability

$$CFL = \max\left(\frac{|u_x|\Delta t}{\Delta x}, \frac{|u_r|\Delta t}{\Delta r}, \frac{|u_\theta|\Delta t}{r\Delta\theta}\right) \lesssim 0.5\text{–}1.0$$

Diffusive stability

$$D = \nu\Delta t\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta r^2} + \frac{1}{(r\Delta\theta)^2}\right) \lesssim 0.5$$

Typical wall-unit time step

$$\Delta t^+ \approx 0.15\text{–}0.4 \quad (Re_\tau = 180)$$

$$\Delta t^+ \approx 0.1\text{–}0.3 \quad (Re_\tau = 395)$$

Implications

- Near wall: small Δr can restrict Δt
- Near centreline: $r\Delta\theta \to 0$ may be restrictive
- Pipe DNS is often more constrained than channel DNS

Key message

Pipe DNS time stepping is affected by both near-wall resolution and cylindrical geometry.

Thermal DNS: Time-Step and Stability Constraints

Diffusion constraints

$$D_\nu = \nu \Delta t \sum_i \frac{1}{\Delta x_i^2}$$

$$D_T = \alpha \Delta t \sum_i \frac{1}{\Delta x_i^2}$$

Key features

- Same mathematical form
- Different coefficients: ν and α
- Relative importance governed by:

$$Pr = \frac{\nu}{\alpha}$$

Prandtl-number regimes

- $Pr > 1$:
  - thermal layer thinner
  - grid resolution may dominate
- $Pr < 1$:
  - $\alpha \gg \nu$
  - thermal diffusion may dominate Δt

Key message

Thermal DNS must satisfy convection, viscous diffusion and thermal diffusion constraints.

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Thermally Developing Flow: Inlet and Setup

Flow characteristics

- Thermal field develops in the streamwise direction
- Hydrodynamic field may already be fully developed
- Velocity and temperature fields are coupled

Inlet treatment

- Specify inlet temperature consistently
- Use fully developed velocity field if possible
- Avoid artificial inlet disturbances

Unheated buffer section

- Often placed before heating/cooling starts
- Helps stabilise the incoming flow
- Reduces abrupt thermal forcing

Potential issue

A sharp transition from unheated to heated wall can introduce artificial transients or instability.

Thermal Boundary Conditions and Property Challenges

Wall thermal boundary conditions

- Constant wall temperature: $T_w = \text{const.}$
- Constant wall heat flux: $q_w = \text{const.}$

Variable-property effects

$$\rho(T), \quad \mu(T), \quad k(T)$$

- Strong variation can introduce numerical stiffness

Discontinuity issue

- At heating onset: $T_w \neq T_{fluid}$
- Sharp wall thermal gradient may produce strong local property gradients

Consequences

- Increased numerical stiffness
- Possible instability or convergence difficulty
- Non-physical oscillations or spikes

Thermal Development: Resolution and Diagnostics

Resolution and domain

- Resolve thermal boundary-layer growth
- High Pr:

$$\Delta y^+_{wall} \lesssim \frac{1}{Pr}$$

- Ensure sufficient streamwise resolution
- Capture thermal entrance length

Key diagnostics

- Bulk temperature: $T_b(x)$
- Nusselt number: $Nu(x)$
- Development towards statistically converged state

Typical issues

Non-smooth $Nu(x)$, incorrect thermal growth rate, or strong sensitivity to inlet/buffer design.

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- Checking DNS Quality

Summary

Runtime Monitoring and Conservation Checks

Conservation checks

- Mass conservation
- Momentum balance
- Thermal energy / heat balance

Monitor during simulation

- CFL number
- Pressure residuals
- Bulk velocity / mass flow rate
- Wall quantities: τ_w, q_w

Probe-based monitoring

- Track signals such as:

$$u(t), \quad T(t)$$

- Use near-wall and centreline locations

What to check

No drift, realistic fluctuations, and no numerical instability.

Statistical Convergence in DNS

Why convergence matters

- DNS fields are time-dependent
- Reliable statistics require sufficient averaging

How to assess

$$\langle U \rangle(t), \qquad \langle u'u' \rangle(t)$$

- Monitor u_τ, bulk velocity and Nu

Typical averaging time

Channel: $50\text{–}100\,T_f$

Pipe: $100\text{–}200\,T_f$

$$T_f = \frac{L_x}{U_b}$$

Key message

Convergence is reached when statistics stop drifting over time.

Key DNS Outputs and Diagnostics

Mean profiles

$$U^+ \text{ vs } y^+$$

$$T^+ \text{ vs } y^+$$

Turbulence statistics

$$\overline{u'u'}, \quad \overline{v'v'}, \quad \overline{w'w'}, \quad \overline{u'v'}$$

Thermal statistics

$$\overline{T'T'}, \qquad \overline{u'T'}$$

Advanced diagnostics

- Budget terms
- Spectra and correlations
- Coherent structures

Key message

DNS provides full-field data for both validation and physics discovery.

Common DNS Mistakes

Resolution and time stepping

- Grid too coarse:
  - missing small scales
  - incorrect stresses
- Time step too large:
  - instability
  - excessive damping

Statistical issues

- Insufficient averaging time
- Initial transient included in statistics

Boundary and output issues

- Inconsistent inlet/outlet or periodic setup
- Sharp thermal boundary-condition jumps
- Output too infrequent for diagnosis

Typical symptoms

Non-physical profiles, noisy statistics, unstable fields, or overly smooth turbulence.

Thermal and Variable-Property Pitfalls

Thermodynamic consistency

- ρ, h and T not updated consistently
- Leads to non-physical energy behaviour

Boundary conditions

- Wrong heat-flux sign
- Incorrect wall-temperature or heat-flux scaling

Thermal resolution

- High-Pr thermal layer under-resolved
- $\Delta y^+_{wall} \lesssim 1/Pr$

Property variation

- $\mu(T)$, $\rho(T)$, $k(T)$ not synchronised
- Strong gradients require smaller Δt

Typical symptoms

Incorrect T^+ or Nu, unphysical temperature fluctuations, or inconsistent wall heat flux.

Practical DNS Grid-Setup Checklist: Physics and Resolution

1. Define flow configuration

- Geometry: channel / pipe / annular
- Physics: isothermal or thermal
- State: fully developed or developing
- Target: Re_τ, Pr

2. Estimate resolution

$$\Delta x^+, \quad \Delta y^+, \quad \Delta z^+$$

Pipe: $\Delta r^+, \quad r\Delta\theta^+$

Near-wall and thermal requirements

$$\Delta y^+_{wall} \lesssim 1$$

$$\Delta y^+_{wall} \lesssim \frac{1}{Pr} \quad \text{if thermal/high-}Pr$$

Developing thermal flow

- Sufficient streamwise resolution
- Domain captures thermal entrance region

Key message

Smallest physical scales set the grid.

Practical DNS Grid-Setup Checklist: Validation

3. Grid quality

- Avoid excessive wall-normal / radial stretching
- Pipe: treat r, θ and centreline carefully

4. Domain adequacy

- Capture large-scale structures
- Use longer domains for thermal development
- Avoid sharp thermal transitions where possible

5. After initial run

- Recompute u_τ
- Update wall-unit spacings
- Verify thermal wall units if applicable

Check constraints

- CFL and diffusion limits
- Mass / energy conservation
- Stable probe signals

Key message

DNS setup is iterative: run, diagnose, refine.

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DNS fundamentals

- Resolves turbulent motions without turbulence modelling
- Captures scales from large eddies to Kolmogorov scales
- Provides reference-quality data at high cost

Stability and convergence

- Time step must satisfy CFL and diffusion constraints
- Thermal and pipe cases can be more restrictive
- Statistical convergence requires sufficient flow-through times

Resolution

- Wall units are essential:

$$\Delta y^+_{wall} \lesssim 1$$

- Pipe flow adds radial/azimuthal and centreline constraints

Diagnostics

- Check conservation, probes, wall quantities and residuals
- Analyse mean profiles, Reynolds stresses and thermal statistics
- Use budgets, spectra and correlations

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